

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe the different methods of wastewater treatment, including primary, secondary and advanced treatments. Discuss the processes involved and the significance of each stage on ensuring clean water.
- Q.24 Evaluate the different methods and equipment available for controlling gaseous and particulate air pollutants. Discuss the principles of operation for technologies as electrostatic precipitators
- Q.25 Write short note on any three:
- a) membrane separation
 - b) absorption by liquid
 - c) fabric filters
 - d) Solid waste

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Roll No.

5th Sem / Chemical (Pulp & Paper)

Subject : Pollution Control in Pulp and Paper Industry

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the main definition of air pollution ?
- a) The presence of harmful substances in the atmosphere
 - b) The presence of natural gases
 - c) The absence of oxygen
 - d) The formation of clouds
- Q.2 Which sampling method collects air quality data at a specific time?
- a) Composite sampling
 - b) Grab sampling
 - c) Ambient sampling
 - d) Continuous Sampling
- Q.3 Which pollutant is classified as a secondary pollutant?

- a) Nitrogen dioxide (NO^2)
- b) Carbon dioxide (CO^2)
- c) Ozone (O^3)
- d) Lead (Pb)

Q.4 What is the main purpose of desulfurization in fuels?

- a) To increase fuel efficiency
- b) To reduce sulfur emissions
- c) To enhance combustion
- d) To reduce the fuel's calorific value

Q.5 What is a common method of solid waste disposal?

- a) Collection b) Incineration
- c) composting d) All of the above

Q.6 Which of the following is a natural source of air pollution?

- a) Vehicle emissions b) Industrial processes
- c) Volcanic eruptions d) Agricultural fertilizers

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 _____ is considered a primary air pollutant?

Q.8 Define water pollution?

Q.9 Name one common sources of air pollution?

Q.10 Name one method of solid waste disposal.

Q.11 BOD is _____

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Q.12 What is the Air Quality Index (AQI)?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 How does air pollution affect human health?

Q.14 What is the difference between recycling and composting?

Q.15 Explain the difference between ambient and stack sampling.

Q.16 Compare the grab sampling and composite sampling techniques, including their advantages and limitations.

Q.17 Analyze the classification of air pollutants according to their origin (primary vs. secondary)

Q.18 Discuss the process of desulfurization and the significance of reducing sulfur dioxide emissions in mitigating air pollution and its health effects.

Q.19 List and explain the methods used for collecting particulate pollutants in the air, with its application.

Q.20 Describe about any one of disposal of solid waste method.

Q.21 Describe about the process like sedimentation, and biological treatment in removing pollutants from wastewater.

Q.22 Describe about the construction and working of spray tower.

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